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1. Introduction to scenario

1.1 Introduction

Taxis have been a part of the transportation sector since the early 1900s when cars were first introduced to society. Now, with the advancement of mobile technology, taxi businesses have incorporated this technology into their business using an app to help users find drivers and drivers find customers. It increases the probability of both finding each other, eases the customer-search for the driver and keeps track of the transactions and bookings taking place throughout the day. It manages all of this and more using a database and a user-friendly interface.

Many e-hailing services are available in Malaysia, each having their advantages and disadvantages. Some have more drivers available, some have cheaper rides and some have more user-friendly softwares.

Many times however, the only drivers available are men. And although companies have strict policies to protect their users from potential harm from their drivers, many women are not comfortable riding with male drivers. Some can be chatty and overbearing which makes many women reluctant to book rides using these apps in the first place.

Our database is for an e-hailing service designed only for women. The drivers can only be women and the customers allowed are women and children below 12 years of age.

The user interface was designed to be easy to use and navigate. The customer can make multiple bookings and view them. Drivers are also able to accept multiple offers and see them.

We also allow the drivers and customers to rate each other and view each other's ratings to help in their decision to choose a driver or accept an offer from a customer.

1.2 Business Rules

1. A customer can have zero or many bookings. Each booking must be associated with one customer.
2. A booking is associated with one car. A car can be used in zero or many bookings.
3. A booking only generates one receipt. A receipt is linked to one booking.
4. A booking can include one or no promotion. A promotion can be applied in zero or many bookings.
5. A car is assigned to one driver. Each driver is assigned to one car.
6. A booking is assigned to one driver through the car. Each driver can have zero or many bookings through their assigned car.

7. A customer can redeem their points for promotions. Promotions are made to be redeemed by customers with sufficient points.
8. A customer can rate multiple drivers. A driver can receive ratings from multiple customers.
9. A driver can rate multiple customers. A customer can receive ratings from multiple drivers.

1.3 User interface and connection with database

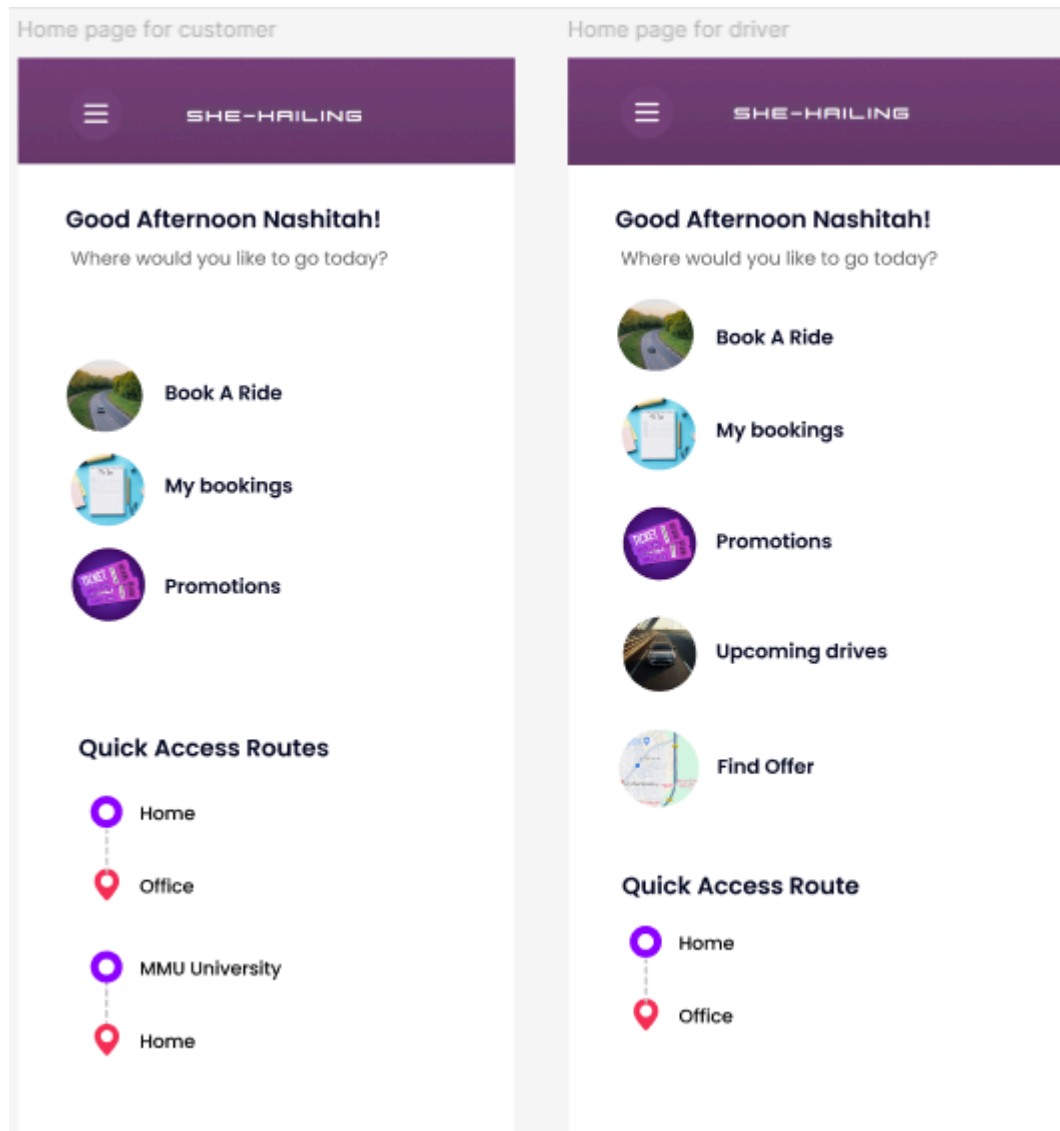


Figure 1: The home page for the customer is shown on the right and the home page for the driver is shown on the left.

“Book A Ride” will create a new row in the Booking table. “My bookings” makes a query to the Booking table using the customer’s Cust_ID. “Promotions” will make a query to the Promotion table using the customer’s Cust_ID. “Upcoming drives” in the driver’s homepage will make a query to the Booking table using the driver’s Driver_ID. Drivers have a Cust_ID and a Driver_ID, which are used to query the Customer table and Driver table respectively depending on what they are using the app for at the moment.

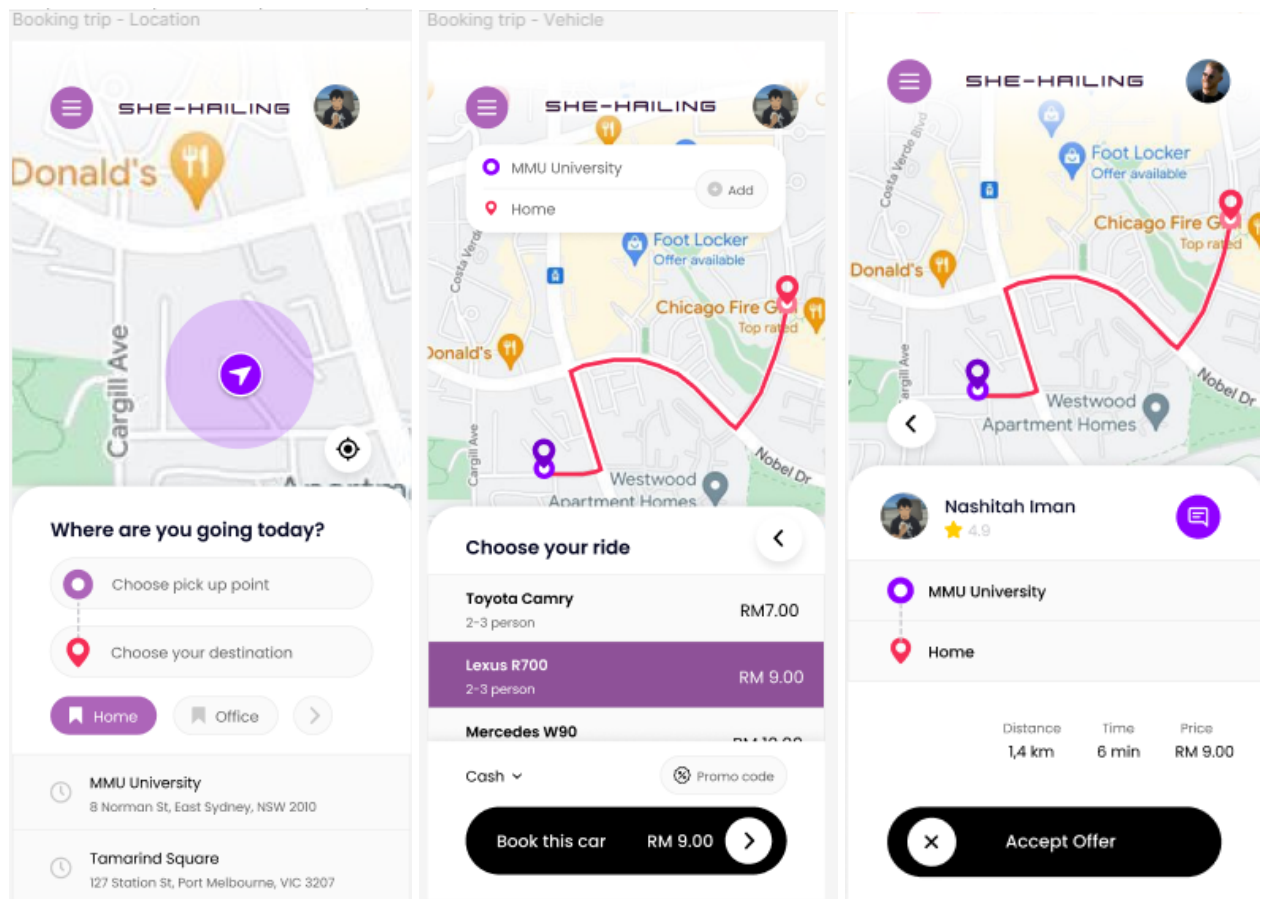


Figure 2: The image on the left shows how the customer chooses the location. The image on the right shows how they choose the vehicle and book the ride. The estimated price of the ride is also shown (later on promotions can be applied). The image on the left is from the driver's point of view. "The Accept Offer" button allows the driver to accept the offer. A message of confirmation will then be sent to the customer.

After the customer chooses the "Book a Ride" option in the main page, he is taken to the booking page where he can type in the location and choose a vehicle. The vehicle is added to the Booking table and is connected to the Car table through the foreign key Car_ID. The location of the customer, the destination and the distance between them is added to the new row in the Booking table.

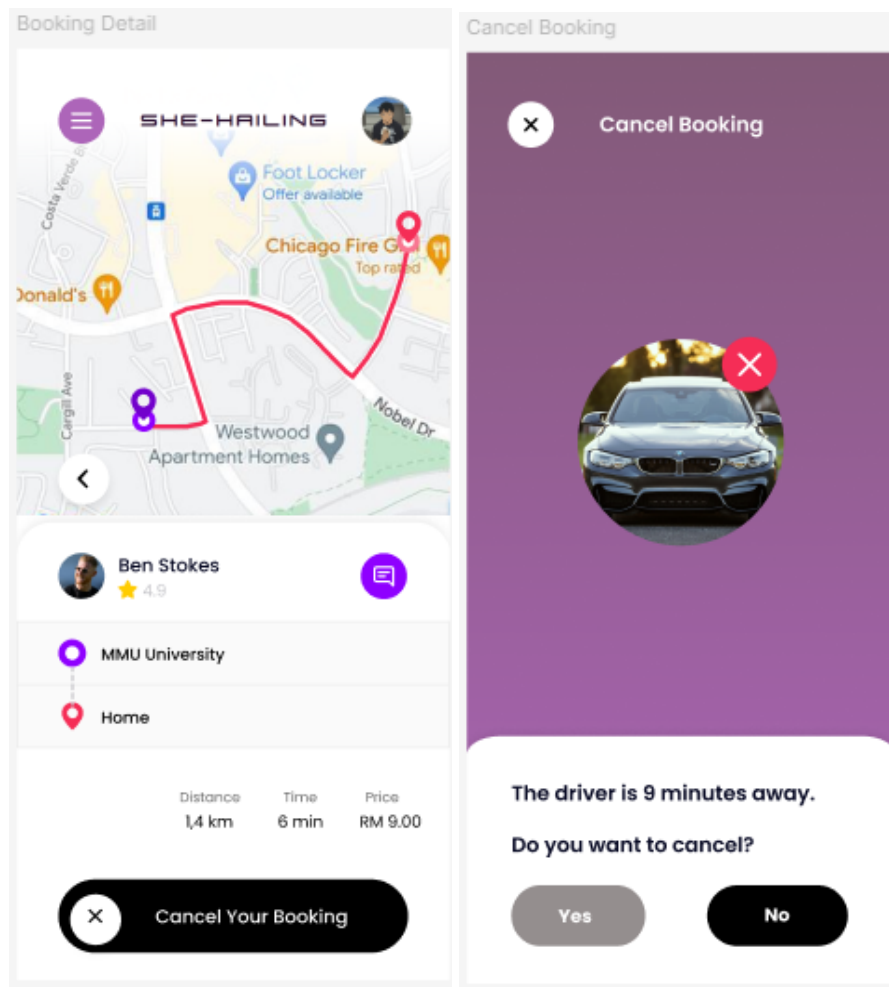


Figure 3: The image on the left shows the booking details once the booking is confirmed. The customer is also allowed to cancel the booking. The image on the right allows the customer to double confirm his cancellation.

If the customer chooses to cancel the trip, the Cancel column in the Booking table will now store “No” instead of “Yes”. The driver will be notified that the booking was canceled. The driver’s rating is retrieved through a query to the Driver_ratings table using the Driver_ID.

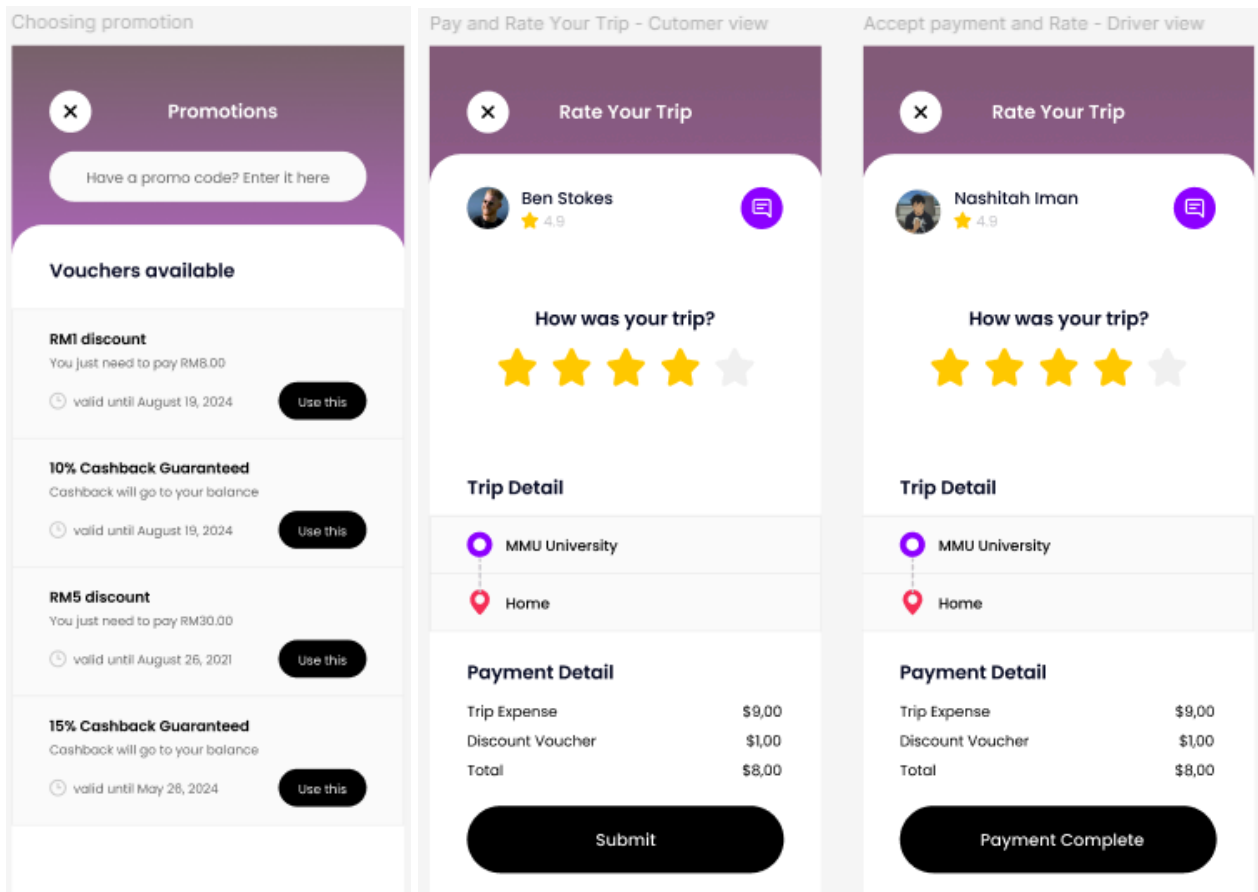


Figure 4: In the first picture from the right, the customer can view their promotions. In the second picture, they are able to rate their drivers and submit payment. The third picture is from the driver's point of view. They are able to rate the customer and confirm the payment has been made.

The promotions are stored in the Promotion table. When the customer clicks "Use this" the Promo_ID is added to the Booking table and Receipt table.

The payment detail shown in the second and third screen in Figure 4 is stored in the Receipt table. The "Trip Expense" is stored under the Price column. The "Discount" is taken from a query made using the Promo_ID in the Receipt table, to find a matching Promo_amount with the same Promo_ID in the Promotion table. The "Total" is stored under the Total Price column in the Receipt table.

The rating by the driver for the is stored in the Cust_ratings table. The rating by the customer for the driver is stored in the Driver_ratings table.

2.0 Scenario overview

2.1 Entities and Attributes (data dictionary)

Receipts								
key	field	Description	Data Type	Size	Constraints	format	Range	FK Reference Table
PK	Receipt_ID	unique id	char	6	Not null	Rxxxxx	RA0000-RZ9999	
FK	Booking_ID	unique issued id	int	8	Not null	XXXX XXXX	00001-99999	Booking table
FK	Promo_ID	unique id	char	6	Not null	Pxxxxx	PA0000-PZ9999	Promotion table
	Price	base price of trip	float	(6,2)	Not null			
	Total price	final price of trip	float	(6,2)	Not null			

Promotion								
key	field	Description	Data Type	Size	Constraints	format	Range	FK Reference Table
PK	Promo_ID	Unique id	char	6	Not null	Pxxxxx	PA0000-PZ9999	
	Promo_type	Type of promotion	varchar	60	Not null			
	Promo_amount	Amount or percentage of the promotion	float	(6,2)	Not Null		0-100	

	Points_ needed	points required get the promotion	int	4	Not null	XXXX	0000-1 000	
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Customer								
key	field	Description	Data Type	Size	Constraints	format	Range	FK Reference Table
PK	Customer_ID	unique id for customers	int	8	Not null	XXXXXXXX	00001-99999	
	Cust_name	customer name	varchar	60	Not null			
	Cust_age	customer age	float	(6,2)	Not Null			
	Cust_phone	customer phone number	int	15	Not null	+xx xx-xxxxxxx		
	Cust_points	customer points	int	4	Not null	XXXX	0000-1000	

Booking								
key	field	Description	Data Type	Size	Constraints	format	Range	FK Reference Table
PK	Booking_ID	unique issued id	int	8	Not null	XXXX XXXX	00001-99999	
	Destination	destination of customer	varchar	200	Not null	XXXX XXXX ...		

	Pickup_loc	pickup location	varchar	200	Not null	XXXX XXXX ...		
	Distance	distance of trip	float	2	Not null	XX		
	Complete	identifier for completion of trip	boolean	1	Not null	X	Y-N	
FK1	Customer_ID	unique id for customers	int	8	Not null	XXXX XXXX	00001-9999	Customer table
FK2	Car_ID	unique id for car	char	8		XXXX XXXX		Car table
FK3	Promo_ID	Unique id	char	6	Not null	PXXX XX	PA0000-P Z9999	promotion table
FK4	Receipt_ID	unique id	char	6	Not null	RXXX XX	RA0000-R Z9999	Receipt table

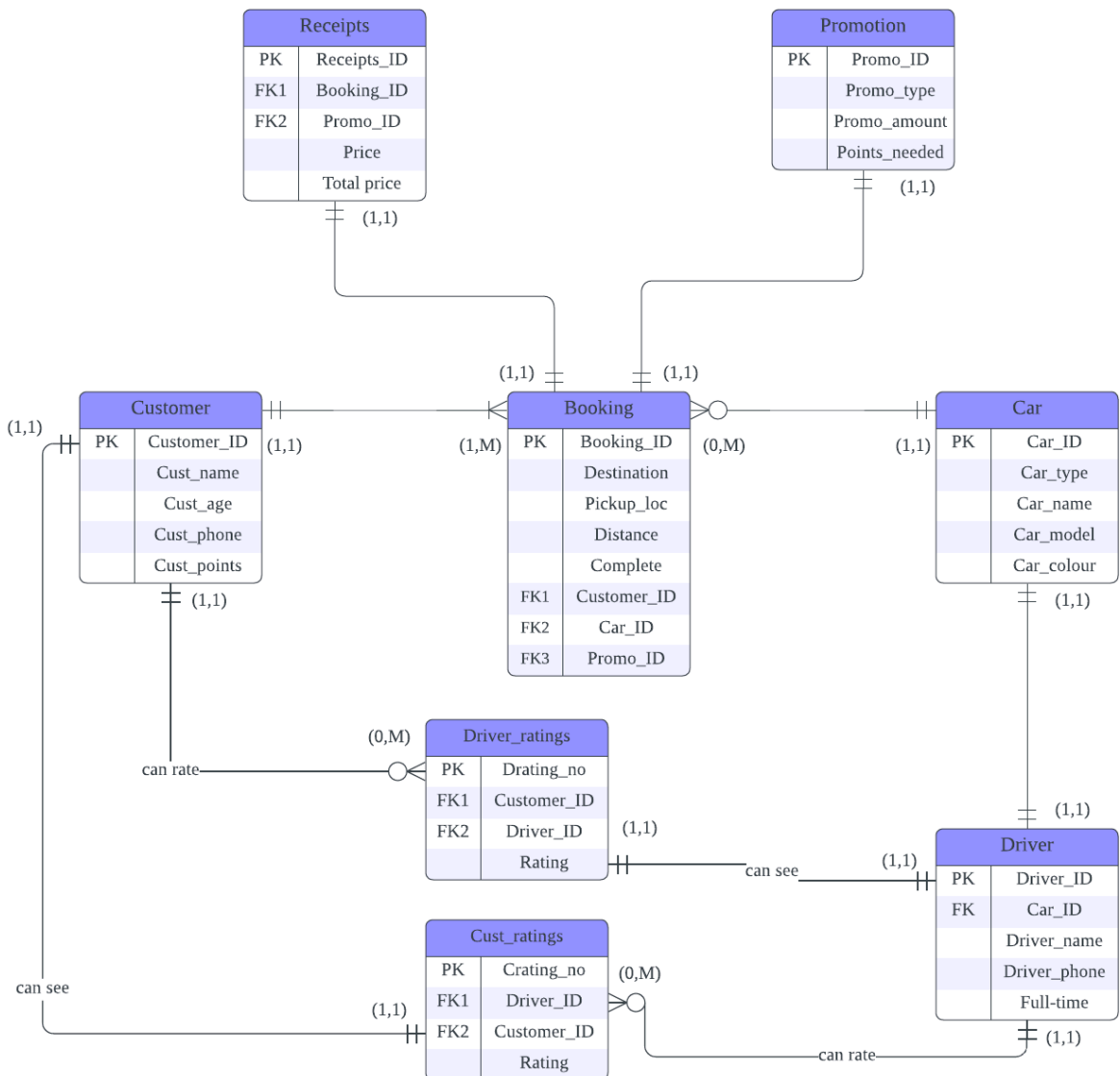
Car								
key	field	Description	Data Type	Size	Constraints	format	Range	FK Reference Table
PK	Car_ID	unique id for car	char	8	Not null	XXXXXXXX X	A00000 -Z9999 9	
	Car_type	type of car	varchar	50	Not null	XXXXXXXX X...		
	Car_name	name of car	varchar	50	Not null	XXXXXXXX X...		
	Car_model	model of car	varchar	40	Not null	XXXXXXXX X...		
	Car_colour	color of car	varchar	15	Not null	XXXXXXXX X...		

Driver_ratings								
key	field	Description	Data Type	Size	Constraints	format	Range	FK Reference Table
PK	Drating_no	driver rating number	int	6	Not null	DRXXXX	DR0000-D99999	
FK1	Customer_ID	unique id for customers	int	8	Not null	XXXXXXXX	00001-9999	Customer table
FK2	Driver_ID	unique id for driver	int	8	Not null	XXXXXXXX	00001-9999	Driver table
	Rating	driver rating	int	1	Not null	X	0-5	

Cust_ratings								
key	field	Description	Data Type	Size	Constraints	format	Range	FK Reference Table
PK	Crating_no	customer rating number	int	6	Not null	CRXXXX	CR0000-CR9999	
FK1	Driver_ID	unique id for driver	int	8	Not null	XXXXXX XX	00001-9999	Driver table
FK2	Customer_ID	unique id for customers	int	8	Not null	XXXXXX XX	00001-9999	Customer table
	Rating	customer rating	int	1	Not null	X	0-5	

Driver								
key	field	Description	Data Type	Size	Constraints	format	Range	FK Reference Table
PK	Driver_ID	unique id for driver	int	8	Not null	XXXXXXXX	00001-99999	
FK	Car_ID	unique id for car	char	8	Not null	XXXXXXXX	A00000-Z99999	Car table
	Driver_name	Driver name	varchar	60	Not null	XXXXXXXX...		
	Driver_phone	Driver phone number	varchar	15	Not null	+xx xx-xxxxxxxx		
	Full-time	does the driver work full-time or not	boolean	1	Not null	X	Y-N	

2.2 Entity Relationship Diagram



2.3 Entity Integrity and Referential Integrity

Referential integrity refers to ensuring the relationships between data within a database are accurate and consistent.

One way to ensure referential integrity is to ensure the primary keys in the tables do not contain NULL values and it contains an “ID” value so that it is identified easily because if primary key contains a null value then we cannot identify those rows. Another way to achieve referential integrity is to ensure there is only one primary key as it should be a unique way to identify data. We also need to make sure that if a foreign key exists in another table it should contain the same exact data as the primary key from the parent table.

We were able to achieve referential integrity in our tables as follow:

1. **Customer:** Each customer will have a unique “Customer_ID” to uniquely identify them from other customers. Since each id is unique, we set it as the primary key. The “Customer_ID” should not be NULL or else it will be difficult to find each customers' information in a database. Since the Customer table does not have foreign keys, it does not need to enforce referential integrity with other tables.. Each field is properly constrained (not null), ensuring that essential information for each promotion is always provided.
2. **Promotion:** Each Promotion will have a unique “Promo_ID” to uniquely identify it from other promotions. Since each id is unique, we set it as the primary key. The “Promo_ID” should not be a NULL value. Since the Promotion table does not have foreign keys, it does not need to enforce referential integrity with other tables. Each field is properly constrained (not null), ensuring that essential information for each promotion is always provided.
3. **Car:** Each Car has a unique “Car_ID” to uniquely identify it from other cars. Since each id is unique, we set it as the primary key. The “Car_ID” should not be a NULL value. Since the Car table does not have foreign keys, it does not need to enforce referential integrity with other tables. Each field is properly constrained (not null), ensuring that essential information for each car is always provided.
4. **Receipts:** The receipts table has 2 Foreign keys “Booking_ID “and “Promo_ID “so All values within both of these fields will be unique since they serve as Primary Key fields in their respective tables. This means, to achieve referential integrity both of these foreign keys have to be the same exact as the primary keys from the respective tables where they come from.
At the same time Each receipt has a unique “Receipt_ID” to uniquely identify it from other Receipts. Since each id is unique, we set it as the primary key. The “receipt_ID” should not be a NULL value. Since the receipt table does have foreign keys, it needs to enforce referential integrity with other tables. Each field is properly constrained (not null), ensuring that essential information for each receipt is always provided.
5. **Booking:** Each Booking has a unique “Booking_ID” to uniquely identify it from other Bookings. Since each id is unique, we set it as the primary key. The “Booking_ID” should not be a NULL value. Since the Booking table does have foreign keys, it

needs to enforce referential integrity with other tables. Each field is properly constrained (not null), ensuring that essential information for each Booking is always provided.

The Booking table has 4 Foreign keys "Customer_ID", "Car_ID", "Receipt_ID" and "Promo_ID" so All values within the 4 of these fields will be unique since they serve as Primary Key fields in their respective tables. This means, to achieve referential integrity all of these foreign keys have to be the same exact as the primary keys from the respective tables where they come from.

6. **Driver_rating:** Each driver rating has a unique "Drating_no" to uniquely identify it from other driver ratings. Since each number is unique, we set it as the primary key. The "Drating_ID" should not be a NULL value. Since the Driver_ratings table does have foreign keys, it needs to enforce referential integrity with other tables. Each field is properly constrained (not null), ensuring that essential information for each rating is always provided.

The Booking table has 2 Foreign keys "Customer_ID" and "Driver_ID" so All values within the 2 of these fields will be unique since they serve as Primary Key fields in their respective tables. This means, to achieve referential integrity both of these foreign keys have to be the same exact as the primary keys from the respective tables where they come from.

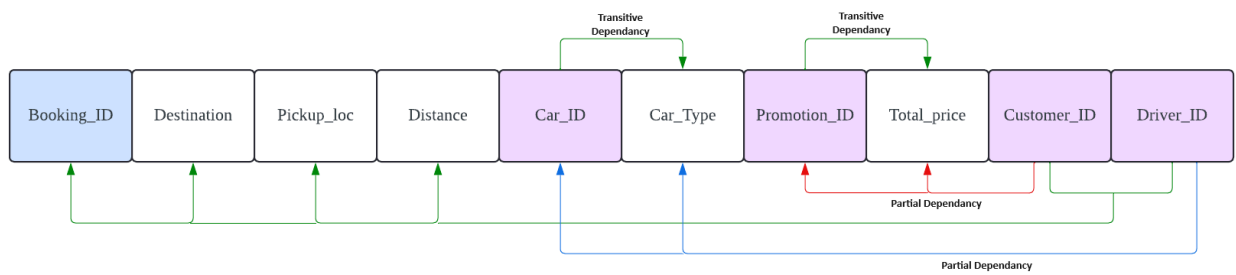
7. **Customer_rating:** Each customer rating has a unique "Crating_no" to uniquely identify it from other customer ratings. Since each number is unique, we set it as the primary key. The "Crating_ID" should not be a NULL value. Since the Customer_ratings table does have foreign keys, it needs to enforce referential integrity with other tables. Each field is properly constrained (not null), ensuring that essential information for each rating is always provided.

The Customer_rating table has 2 Foreign keys "Customer_ID" and "Driver_ID" so All values within the 2 of these fields will be unique since they serve as Primary Key fields in their respective tables which are the customer table and the driver table. This means, to achieve referential integrity both of these foreign keys have to be the same exact as the primary keys from the respective tables where they come from.

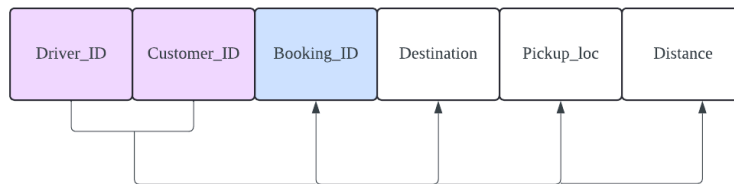
8. **Driver:** Each driver has a unique "Driver_ID" to uniquely identify it from other drivers. Since each id is unique, we set it as the primary key. The "Driver_ID" should not be a NULL value. Since the driver table does have foreign keys, it needs to enforce referential integrity with other tables. Each field is properly constrained (not null), ensuring that essential information for each driver is always provided.

The driver table has 1 Foreign key "Car_ID" so the values in this field will be unique since it serves as a Primary Key field in its respective table which is the Car table. This means, to achieve referential integrity this foreign key has to have the same exact value as the primary key from the car table.

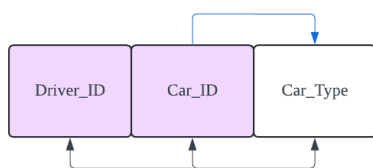
3. Normalization



3rd Normal Form



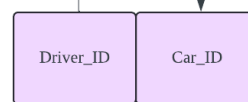
2nd Normal Form



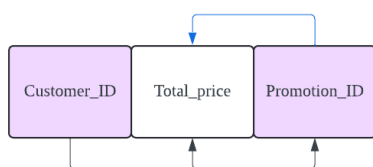
Removing transitive dependency



3rd Normal Form



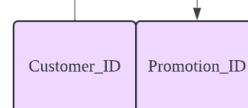
2nd Normal Form



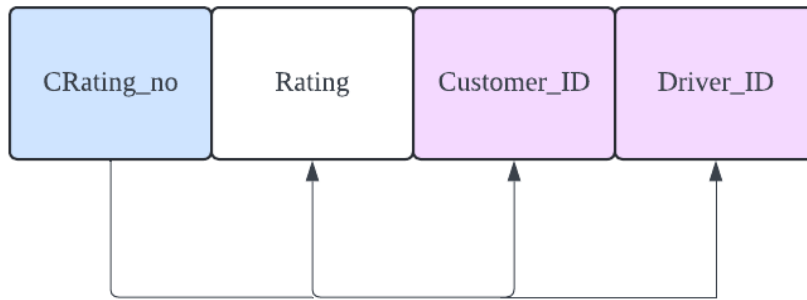
Removing transitive dependency



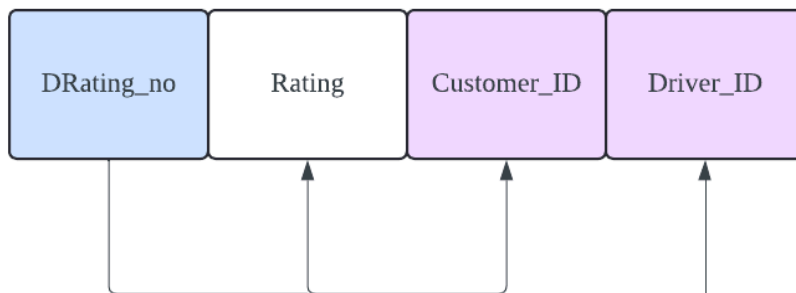
3rd Normal Form



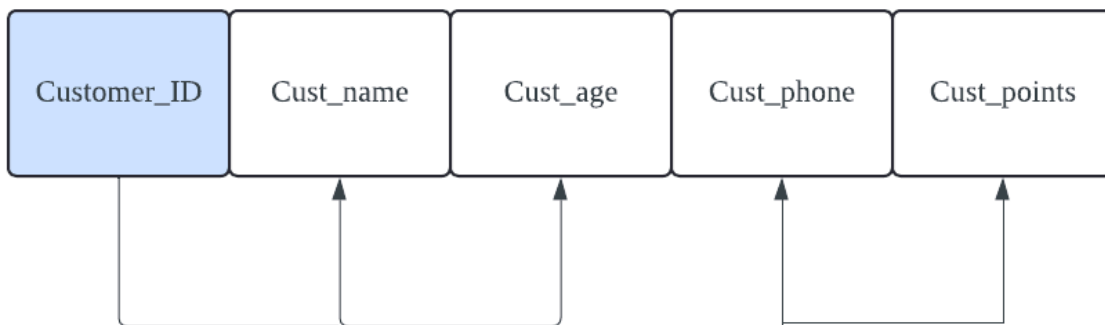
The Booking table was in 1st normal form. The partial dependencies were broken and three tables were created in its place. The first has no transitive dependencies so it is already in 3rd normal form. It is used as the Booking Table. The other two 2nd normal form tables had their transitive dependencies broken. Four 3rd normal form tables were made: the Driver table, the Car table, and the Receipt table. The table with Customer_ID and Promotion_ID was not used further as it was redundant in the system.



Cust_ratings Table



Driver_ratings Table



Customer Table

The three tables Customer, Cust_ratings, Driver_ratings, do not have any partial or transitive dependencies. Therefore, they are already in 3rd normal form and do not need to be simplified further.

4. Data Dictionary

Customer Table

<i>Customer_ID</i>	<i>Cust_name</i>	<i>Cust_age</i>	<i>Cust_phone</i>	<i>Cust_points</i>
1	Willie Rose	25	0112551489	58
2	Magnolia Arias	24	0132561847	45
3	Alec Hopkins	35	0152489567	74
4	Gabriela Carson	47	0162548795	125
5	Ares Pollard	85	0125478963	36
6	Alice Park	32	0147890867	104
7	Alexandra Mleux	21	0157803539	87
8	Nur Fatimah	55	0178964263	67
9	Hana Humaira	24	0198632572	34
10	Roseanne Park	27	0178265375	150
11	Jennie Kim	28	0154789865	49
12	Marissa Iman	16	01165673009	11
13	Havana Liu	23	0176540910	53
14	Julie Han	24	0134421610	2
15	Priya Kumar	60	0149140444	66
16	John doe	58	0116879945	99
17	Ashley Graham	20	0165563201	28
18	Goro takuma	42	0182094857	76
19	Mina Mikes	34	0115944738	12
20	Elizabeth bell	63	0129384755	130

Promotion Table

<i>Promo_ID</i>	<i>Promo_type</i>	<i>Promo_amount</i>	<i>Points_needed</i>
PA0589	RM1 off	1.00	50
PA2031	RM5 off	5.00	100
PA4305	RM10 off	10.00	150
PA2405	RM15 off	15.00	200
PA3505	RM20 off	20.00	350
PA0001	RM0	0.00	0
PA2104	RM3 off	3.00	80
PA2603	RM25 off	25.00	400
PA8506	RM8 off	8.00	120
PA2604	RM12 off	12.00	160
PA8507	RM18 Off	18.00	240
PA1103	RM0.20 off	0.20	20
PA2203	RM0.40 off	0.40	40
PA1303	RM0.60 off	0.60	60
PA4403	RM0.80 off	0.80	80

PA0503	RM0.10 off	0.10	100
PA0511	RM0.70 off	0.70	70
PA0531	RM2 off	2.00	75
PA2051	RM28 off	28.00	380
PA1891	RM16	16.00	220

Car table

Car_ID	Car_Type	Car_name	Car_model
A4325602	4-pax	Proton	Saga
A7685748	7-pax	Perodua	Alza
A0975982	7-pax	Proton	Exora
A7891456	4-pax	Perodua	Myvi
A2348745	4-pax	Perodua	Axia
A3451234	4-pax	Toyota	Yaris
A0876789	7-pax	Kia	Carnival
A1283078	4-pax	Mercedes Benz	C-Class
A7061145	7-pax	Mitsubishi	Xpander
A2562034	4-pax	Honda	Civic
A7524487	4-pax	Perodua	Bezza
A0489605	7-pax	Toyota	Vellfire
A1713438	4-pax	Hyundai	Elantra
A8712319	4-pax	Toyota	Prius
A0879400	2-pax	Perodua	Viva
A1168125	4-pax	Proton	Iriz
A8696315	4-pax	Lexus	ES
A1269856	7-pax	Toyota	Highlander
A1233254	7-pax	Perodua	Aruz
A7892145	4-pax	Kia	Cerato

Driver table

Driver_ID	Car_ID	Driver_name	Driver_phone	Full_time
1	A4325602	Felicity Hunt	0125648796	Yes
2	A7685748	Juan Leonard	0152684789	No
3	A0975982	Addilynn Salas	0112567894	Yes
4	A7891456	Zaiden Baker	0135478965	Yes
5	A2348745	Isla Cabrera	0125486971	Yes
6	A3451234	Isabella Haris	0134786392	No
7	A0876789	Nadia Omar	0198724665	Yes
8	A1283078	Sarah Holland	0129374673	No

9	A7061145	Priyanka Chopra	0198767354	No
10	A2562034	Sofea Alyssa	0113267629	Yes
11	A0879400	Siti Zainab	0156564909	Yes
12	A1713438	Umi Halija	0141198505	No
13	A8712319	Phoebe Bridgers	0120240906	Yes
14	A7524487	Ethel Cain	0131445353	No
15	A0489605	Sarah Chow	0182273234	Yes
16	A1168125	Leila Mansoor	0118930499	Yes
17	A8696315	Siti Khadijah	0123984074	Yes
18	A1269856	Jing Yu	0192036048	No
19	A1233254	Adya Mahendran	0159284702	Yes
20	A7892145	Amira Yousef	0112937049	No

Booking Table

<i>Booking_ID</i>	<i>Destination</i>	<i>Pickup_loc</i>	<i>Distance(km)</i>	<i>Complete</i>	<i>Customer_ID</i>	<i>Car_ID</i>	<i>Promo_ID</i>
200	Masjid Putra	MMU, Cyberjaya	15	Yes	1	A4325602	PA0589
201	Suria KLCC	Metropoint, Kajang	30	No	2	A7685748	PA0001
202	Lotuss, Ampang	Suria KLCC	15	Yes	3	A0975982	PA0589
203	IOI Putrajaya	PV8, Gombak	40	Yes	4	A7891456	PA2031
204	Exhaustion, Damansara	Happiness, Putrajaya	35	No	5	A2348745	PA0001
205	TRX Exchange	UOC Cyberjaya	35	Yes	6	A3451234	PA2104
206	Alamanda Mall	MRT station Cyberjaya Utara	20	No	7	A0876789	PA2104
207	Taman Saujana Hijau	Neocyber	16	No	8	A1283078	PA0589
208	Bukit Bintang	MMU Cyberjaya	36	Yes	9	A7061145	PA8506
209	KLIA Sepang	Kota Warisan	25	Yes	10	A2562034	PA2031
210	Lalaport	TRX Exchange	5	Yes	11	A7524487	PA2104
211	MMU Cyberjaya	SS15, Subang Jaya	30	No	12	A0489605	PA0001
212	The Five,	Alam Damai	20	No	13	A1713438	PA2203

	Bukit Damansara						
213	Jalan Tunku Abdul Rahman	Bangi Central	27	Yes	14	A8712319	PA1303
214	Ampang Point	Stadium Bukit Jalil	11	Yes	15	A0879400	PA2203
215	Dpulze Mall	Seria Residence,Cyberjaya	6	Yes	16	A1168125	PA2203
216	Suria KLCC	Tamarind Square,Cyberjaya	32	Yes	17	A8696315	PA4305
217	KI Sentral Bus station	Berjaya Times square ,KL	20	Yes	18	A1269856	PA0503
218	Sunway Pyramid	KI Sentral	22	No	19	A1233254	PA1303
219	Swiss Hotel	Klcc Park, kl	9	No	20	A7892145	PA0589
220	Aeon Shah Alam	Shah Alam	18	Yes	15	A7524487	PA2104
221	Wangsa Walk Mall	Wangsa Maju	15	No	6	A1713438	PA2203
222	Quill City Mall	Kampung Baru	8	No	1	A1168125	PA2203
223	Sungei Wang Plaza	Bukit Bintang	5	Yes	20	A8696315	PA4305
224	Paradigm Wang Plaza	Kelana Jaya	15	No	14	A1269856	PA0503
225	Lot 10	Bukit Bintang	8	No	3	A7892145	PA0589
226	The Gardens Mall	Mid Valley City	7	Yes	5	A2348745	PA4305
227	Central Market	Kampung Baru	1	Yes	13	A1713438	PA2203
228	KL Tower	Bukit Nanas	9	Yes	16	A8712319	PA1303
230	Empire Shopping Gallery	Subang Jaya	20	Yes	4	A1168125	PA2203

Receipts Table

Receipt_ID	Booking_ID	Promo_ID	Price	Total_Price
RA0401	200	PA0589	7.5	6.5
RA0402	201	PA0001	15	15
RA0403	202	PA0589	7.5	6.5
RA0404	203	PA2031	20	15
RA0405	204	PA0001	17.5	17.5

RA0406	205	PA2104	17.5	14.5
RA0407	206	PA2104	10	7
RA0408	207	PA0589	8	7
RA0409	208	PA8506	18	10
RA0410	209	PA2031	12.5	7.5
RA0411	210	PA2104	15	15
RA0412	211	PA0001	13.5	12.9
RA0413	212	PA2203	2.5	2.4
RA0414	213	PA1303	5.5	5.1
RA0415	214	PA2203	10	9.6
RA0416	215	PA2203	5	4.6
RA0417	216	PA4305	20.5	10.5
RA0418	217	PA0503	10	9.9
RA0419	218	PA1303	12.5	11.9
RA0420	219	PA0589	7	6
RA0421	220	PA2104	11	8
RA0422	221	PA2203	16	15.6
RA0423	222	PA2203	19	18.6
RA0424	223	PA4305	20	10
RA0425	224	PA0503	5	4.9
RA0426	225	PA0589	6	5
RA0427	226	PA1303	7	6.4
RA0428	227	PA2203	8	7.6
RA0429	228	PA1303	12	11.4
RA0430	230	PA2203	15	14.6

Driver_ratings table

<i>Drating_no</i>	<i>Customer_ID</i>	<i>Driver_ID</i>	<i>Rating</i>
DR0001	1	1	5
DR0002	2	2	4
DR0003	3	3	3
DR0004	4	4	4
DR0005	5	5	5
DR0006	6	6	4
DR0007	7	7	5
DR0008	8	8	5
DR0009	9	9	3
DR0010	10	10	4
DR0011	11	11	5
DR0012	12	12	4
DR0013	13	13	5
DR0014	14	14	2
DR0015	15	15	3
DR0016	16	16	5

DR0017	17	17	4
DR0018	18	18	5
DR0019	19	19	3
DR0020	20	20	5

Cust_ratings table

Crating_no	Driver_ID	Customer_ID	Rating
CR0001	1	1	4
CR0002	2	2	5
CR0003	3	3	3
CR0004	4	4	4
CR0005	5	5	5
CR0006	6	6	3
CR0007	7	7	3
CR0008	8	8	5
CR0009	9	9	4
CR0010	10	10	5
CR0011	11	11	5
CR0012	12	12	5
CR0013	13	13	4
CR0014	14	14	1
CR0015	15	15	5
CR0016	16	16	5
CR0017	17	17	3
CR0018	18	18	5
CR0019	19	19	4
CR0020	20	20	5